

Комплексные числа

$$\mathbb{C} = \{(a, b) \mid a, b \in \mathbb{R}\}$$

$$(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2)$$

$$(a_1, b_1)(a_2, b_2) = (a_1 a_2 - b_1 b_2, a_1 b_2 + a_2 b_1)$$

$$(4, -5) + (2, 5) \cdot (-7, 7) = (4, -3) + \begin{pmatrix} 2 & -1 - 5 \cdot 7 \\ 2 & 7 + 5 \cdot (-7) \end{pmatrix} \\ = (4, -3) + (-37, 9) = (-33, 6)$$

* $\boxed{6}$ Пусть $z_j = (a_j, b_j) \in \mathbb{C}, j = 1, 2, 3$

$$1 \quad z_1 + z_2 = z_2 + z_1$$

$$2 \quad (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

$$3 \quad z_1 + (0, 0) = z_1$$

$$4 \quad (a, b) + (-a, -b) = 0, 0$$

$$(-a, -b) = -(a, b)$$

$$z_2 - z_1 = z_2 + (-z_1)$$

$$5 \quad z_1 z_2 = z_2 z_1$$

$$6 \quad (z_1 z_2) z_3 = z_1 (z_2 z_3)$$

$$7 \quad z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$$

$$(a_1, b_1) \cdot ((a_2, b_2) + (a_3, b_3)) = (a_1, b_1) \cdot (a_2 + a_3, b_2 + b_3) = (a_1, (a_2 + a_3) - b_1, (b_2 + b_3) + b_1(a_2 + a_3))$$

$$= (a_1 a_2 + a_1 a_3 - b_1 b_2 - b_1 b_3, a_1 b_2 + a_1 b_3 + b_1 a_2 + b_1 a_3)$$

$$= (a_1 a_2 - b_1 b_2, a_1 b_2 + b_1 a_2) + (a_1 a_3 - b_1 b_3, a_1 b_3 + b_1 a_3)$$

$$= (a_1, b_1) (a_2, b_2) + (a_1, b_1) (a_3, b_3)$$

$$\text{Зг } \mathbb{R}$$

$$1 \quad a + b = b + a$$

$$2 \quad a + (b + c) = (a + b) + c$$

$$3 \quad a + 0 = 0, \forall a$$

$$4 \quad \forall a \in \mathbb{R} \exists b \in \mathbb{R} \\ a + b = 0, \quad b = -a$$

$$5 \quad a \cdot b = b \cdot a$$

$$6 \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$$

$$7 \quad a \cdot (b + c) = ab + ac \quad \forall a, b, c$$

$$8 \quad a \cdot 1 = a \quad \forall a$$

$$9 \quad \forall a \neq 0 \exists c \\ ac = 1, \quad c = \frac{1}{a}$$

$$\mathcal{F}(a, b)(?, 0) = a, b /$$

$$(a, 1 - b, 0, a, 0 + b, 1)$$

$$\tilde{K} = \{(a, 0) \mid a \in \mathbb{R}\} \subset \mathbb{C}$$

$$(a, 0) + (c, 0) = (a+c, 0) \in \tilde{K}$$

$$(a, 0)(c, 0) = (ac, 0) \in \tilde{K}$$

$$a = (a, 0)$$

$$\| \tilde{K} = \tilde{K} \subset \mathbb{C}$$

$$a(b, c) = (a, 0)(b, c) = (a, 0)(b, c) = (ab, ac)$$

$$(a, b) = (a, 0) + (0, b) = (a, 0) + (b, 0)(0, 1) = a + b \cdot i$$

$$i^2 = (0, 1)(0, 1) = (0-1, 0+0) = (-1, 0)$$

$$(6+5i)(-7+i) = 6(-7) + 6i + 5i(-7) + 5i^2 =$$

$$= -42 + 6i - 35i - 5 = -47 - 29i$$

$$(2+i)^4 = 2^4 + 4 \cdot 2^3 i + 6 \cdot 2^2 i^2 + 4 \cdot 2 i^3 + i^4 =$$

$$= (16 - 32i - 24 + 8i + 1) = -7 - 24i$$

$$z = (a, b) = a + bi \quad \bar{z} = (a, -b) = a - bi \quad \overline{(5 - 7i)} = 5 + 7i$$

$$\overline{-2 + 3i} = -2 - 3i$$

IB) 1 $\bar{\bar{z}} = z$

2 $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

3 $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$

4 $z \cdot \bar{z} = 2 \operatorname{Re} z$

5 $z \cdot \bar{z} = (a + bi)(a - bi) = a^2 - b^2 i^2 = a^2 + b^2 = |z|^2 \in \mathbb{R}$

$$\overline{z_1 \cdot z_2} = (a_1 - bi_1)(a_2 - bi_2) = a_1 a_2 - a_1 b_2 i + b_1 a_2 i - b_1 b_2 i^2 = (a_1 a_2 + b_1 b_2) - (a_1 b_2 - a_2 b_1)i =$$

$$\sqrt{z \bar{z}} = |z| = \sqrt{a^2 + b^2} = |a + bi| / |a + 0i| = \sqrt{a^2} = |a|$$

$$\overline{(a_1 + bi_1)(a_2 + bi_2)} = \overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$$

Кстати $z \neq 0 \Rightarrow z \cdot \bar{z} = |z|^2 \neq 0 \in \mathbb{R}$

$\Rightarrow \exists (|z|^2)^{-1} = \frac{1}{|z|^2}$

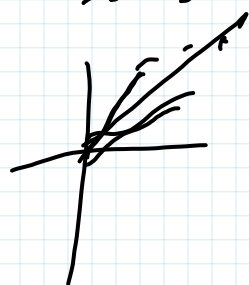
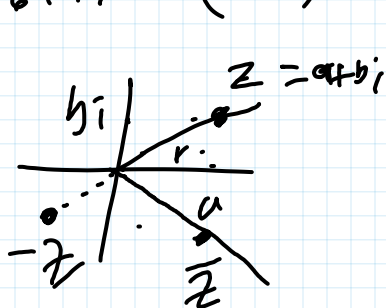
$$z \cdot \bar{z} \cdot \frac{1}{|z|^2} = 1 \Rightarrow z \cdot \frac{\bar{z}}{|z|^2} = 1$$

$$\frac{1}{|z|^2} = \frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

$$(2 + 3i)^{-1} = \frac{1}{2 + 3i} = \frac{1 \cdot (2 - 3i)}{(2 + 3i)(2 - 3i)} =$$

$$= \frac{2 - 3i}{4 + 9} = \frac{2}{13} - \frac{3}{13}i$$

$$\frac{5 - 4i}{6 + 7i} = \frac{5 - 4i}{(6 + 7i)(6 - 7i)} = \frac{30 - 35i - 26i + 28i^2}{36 + 49} = \frac{2}{85} - \frac{59}{85}i$$



$$\sqrt{a^2 + b^2} = |z| = r$$

$$a = r \cos \varphi \quad \cos \varphi = \frac{a}{r}$$

$$b = r \sin \varphi \quad \sin \varphi = \frac{b}{r}$$

$$z = a + bi = r(\cos \varphi + i \sin \varphi)$$

числото 0 е еквивалентно със трети въз

$$\begin{aligned} & r_1(\cos \varphi_1 + i \sin \varphi_1) r_2(\cos \varphi_2 + i \sin \varphi_2) = r_1 r_2 (\cos \varphi_1 \cos \varphi_2 - \sin \varphi_1 \sin \varphi_2 + i(\sin \varphi_1 \cos \varphi_2 + \cos \varphi_1 \sin \varphi_2)) \\ & = r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2)) = 1(\cos 120 + i \sin 120) = \cos 75 + i \sin 75 \end{aligned}$$

$$T \quad \frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{r_1 r_2 (\cos \varphi_1 + i \sin \varphi_1) (\cos \varphi_2 - i \sin \varphi_2)}{r_2^2} = \frac{r_1 r_2 (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2))}{r_2^2}$$

~~*~~ формула на Мавер

$$T \quad (r(\cos \varphi + i \sin \varphi))^n = r^n (\cos(n\varphi) + i \sin(n\varphi))$$

$$? n=1 \quad r(\cos \varphi + i \sin \varphi) = r^1 (\cos \varphi \cdot 1 + i \sin \varphi)$$

2 Предполагаме че е вярно за n

3 Проб за $n+1$

$$\begin{aligned} [r(\cos \varphi + i \sin \varphi)]^{n+1} &= [r(\cos \varphi + i \sin \varphi)]^n \cdot r(\cos \varphi + i \sin \varphi) = \\ &= r^n (\cos n\varphi + i \sin n\varphi) \cdot r(\cos \varphi + i \sin \varphi) = r^{n+1} (\cos(n\varphi + \varphi) + i \sin(n\varphi + \varphi)) = \\ &= r^{n+1} (\cos(n+1)\varphi + i \sin(n+1)\varphi) \end{aligned}$$

$\Rightarrow$ е изгнано за $\forall n \in \mathbb{N}$

T Нека $z = r(\cos \varphi + i \sin \varphi) \neq 0$

$x^n = z$ има n решения

n -ти корен на z

$$\sqrt[n]{z} = \left\{ \sqrt[n]{r} \left(\cos \frac{\varphi + 2k\pi}{n} + i \sin \frac{\varphi + 2k\pi}{n} \right) \right\}_{k=0,1,\dots,n-1}$$

$$\begin{aligned} & t^n = z \quad t = \rho(\cos \alpha + i \sin \alpha) \\ & t^n = z = \rho^n (\cos n\alpha + i \sin n\alpha) \end{aligned}$$

$$\rho = \sqrt[n]{r} \quad \text{и} \quad \alpha = \varphi + 2k\pi \Rightarrow \alpha = \frac{\varphi + 2k\pi}{n}$$

Проверка

$$\left(\sqrt[n]{r} \cos \frac{\varphi + 2k\pi}{n} + i \sin \frac{\varphi + 2k\pi}{n} \right)^n =$$

$$= \sqrt[n]{r^n} \left(\cos \frac{\varphi + 2k\pi}{n} + i \sin \frac{\varphi + 2k\pi}{n} \right)^n$$

$$\text{Нека } w_k = \sqrt[n]{r} \dots$$

$$w_k \stackrel{?}{=} w_s$$

$$\frac{n + 2\pi k}{n} - \frac{\varphi + 2\pi s}{n} = 2\pi l \quad l \in \mathbb{Z}$$

$$2\pi k - 2\pi s = 2\pi l \quad | : 2\pi$$

$$k - s = l \Leftrightarrow k = s + l \quad k, s, l \in \mathbb{Z}$$

при нѣ брѣм крѣм и л з

Реш

* Опр Нека $F \neq \emptyset$, с опре $+$ и $\cdot$

$$\text{т.е. } \forall a, b \in F \quad a + b \in F$$

$$a \cdot b \in F$$



Кратко мерно векторно пространство и минимална система

$$F \neq \emptyset, +, \cdot$$

$$\forall a, b \in F \begin{cases} a+b \in F \\ a \cdot b \in F \end{cases}$$

F е поле ако

$$\begin{aligned} 1 & a+b = b+a \\ 2 & (a+b)+c = a+(b+c) \\ 3 & \exists 0 \in F, a+0=a, \forall a \\ 4 & \forall a \in F \exists b \in F: \\ & \quad a+b=0 \end{aligned}$$

$$\begin{aligned} 5 & ab = ba \\ 6 & a(bc) = (ab)c \\ 7 & a(b+c) = ab+ac \\ 8 & \exists 1 \in F, a \cdot 1 = a \\ 9 & \forall a \neq 0 \exists c \in F \\ & \quad a \cdot c = 1 \end{aligned}$$

F^n - n -мерно векторно пространство

$$\begin{aligned} a &= (a_1, \dots, a_n) \\ b &= (b_1, \dots, b_n) \end{aligned} \Rightarrow a+b = (a_1+b_1, \dots, a_n+b_n)$$

$$\forall d \in F, \forall a \in F^n$$

$$da = (da_1, da_2, \dots)$$

$$2a + 3b = 2(1, 2, 3) + 3(2, 3, 1) = (2, 4, 6) + (6, 9, 3) = (8, 13, 9)$$

* TE $a = (a_1, \dots, a_n), b = (b_1, \dots, b_n) \in F^n$

$$1 \quad a+b = b+a$$

$$2 \quad (a+b)+c = a+(b+c)$$

$$3 \quad a+(0, 0, \dots, 0) = a \quad \Theta = (0, 0, \dots, 0)$$

$$4 \quad a+(-a) = \Theta$$

$$(a_1, \dots, a_n) + (-a_1, \dots, -a_n)$$

$$5 \quad 1 \in F \text{ и } da = a \quad \forall a \in F^n$$

$$6 \quad d(a+b) = da + db \quad \forall d \in F$$

$$7 \quad (d+e)a = da + ea$$

$$8 \quad (d \cdot e)a = d(ea)$$

Това са само основните свойства

$$\begin{aligned}
 (1+\varphi) \cdot &= (1+\varphi) \cdot (a_1, a_2, \dots, a_n) = ((1+\varphi)a_1, (1+\varphi)a_2, \dots) \\
 &= (1a_1, 1a_2, \dots, 1a_n) + (\varphi a_1, \varphi a_2, \dots, \varphi a_n) = 1(a_1, a_2, \dots) + \varphi(a_1, a_2, \dots) = \\
 &= 1a + \varphi a \\
 \boxed{1(a_1, a_n) = 1a_1, 1a_n = (a_1, a_n)}
 \end{aligned}$$

* Те. $1a, k \in F^n$ — это $a+x=b$ или единственное решение

$$x = b + (-a) = b - a$$

$$a + (b + (-a)) = a + ((-a) + b) = (a + (-a)) + b = 0 + b = b$$

$$x = b + (-a) \in \text{реш.}$$

$$\begin{aligned}
 \text{Нес. } y \in \text{реш. } a + y = b \\
 a + y + (-a) = b + (-a) \\
 a + (-a) + y = b + (-a) \\
 0 + y = b + (-a) \\
 y = b + (-a)
 \end{aligned}
 \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{единственно} \\ \text{реш.} \end{array}$$

$$(5+3i, 4-2i, 7+9i) + (7+i)x = (19+12i, 5-2i, 8+7i)$$

$$(7+i)x = (19+12i, 5-2i, 8+7i) - (5+3i, 4-2i, 7+9i)$$

$$(7+i)x = (14+9i, 1, 7+3i) / \left(\frac{7}{2} - \frac{1}{2}i\right)$$

$$x = \left((14+9i) \left(\frac{1-i}{2}\right), 1 \left(\frac{1-i}{2}\right), (7+3i) \left(\frac{1-i}{2}\right) \right)$$

$$x = \left(\frac{14-9}{2} + \frac{9-14}{2}i, \frac{1-i}{2}, \frac{1+3}{2} + \frac{-7+3}{2}i \right)$$

$$x = \left(\frac{5}{2} - \frac{5}{2}i, \frac{1-i}{2}, 2+i \right)$$

$$\begin{aligned}
 (7+i)^{-1} &= \frac{1}{7+i} = \frac{7-i}{(7+i)(7-i)} = \\
 &= \frac{7-i}{2} = \frac{7}{2} - \frac{1}{2}i
 \end{aligned}$$

$$\text{Сб-б. } \lambda a = 0 \Leftrightarrow \lambda = 0 \text{ или } a = 0$$

$$\Leftrightarrow 0a = 0; \lambda 0 = 0$$

$$\Rightarrow \text{Нес. } \lambda a = 0 \text{ и } \lambda \in F_n \text{ нес. } \lambda \neq 0 \Rightarrow \exists \lambda^{-1} \in F$$

$$\lambda a = 0 / \lambda^{-1} = \underbrace{\lambda^{-1} \lambda}_1 a = \underbrace{\lambda^{-1} 0}_0 = 0 \Rightarrow a = 0$$

2 6-6a . . .

$\alpha = (\alpha, 2\alpha, 3\alpha, 4\alpha)$ $\alpha = ?$ d е мн конст на α, b, c

$$d = \lambda a + \varphi b + \psi d$$

$$\lambda(3, 5, 7, 1) + \varphi(2, -1, 4, 0) + \psi(1, 2, 0, -1) = (\alpha, 2\alpha, 3\alpha, 4\alpha)$$

$$(3\lambda + 2\varphi + \psi, 5\lambda - \varphi + 2\psi, 7\lambda + 4\varphi, \lambda - \psi) = (\alpha, 2\alpha, 3\alpha, 4\alpha)$$

$$\begin{cases} 3\lambda + 2\varphi + \psi = \alpha \\ 5\lambda - \varphi + 2\psi = 2\alpha \\ 7\lambda + 4\varphi = 3\alpha \\ \lambda - \psi = 4\alpha \end{cases}$$

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b$$

$$a_i \in F \quad b \in F$$

$x_1, x_2, \dots, x_n$ неизвестни

$(\beta_1, \beta_2, \dots, \beta_n) \in F^n$ е решение

на q -это

$$a_1 \beta_1 + a_2 \beta_2 + \dots + a_n \beta_n = b$$

е т.е. β е решение

(т.е. β е решение)

$$\forall T \in \mathcal{T}, x_1 + a_n x_n = b$$

1 когато $a_1 = a_2 = \dots = a_n = 0$

Всички вектори от F^n е при

2 $a_1 = a_2 = 0$ б.т.0 не е решение

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{k1}x_1 + \dots + a_{kn}x_n = b_k \end{cases}$$

$$(\beta_1, \dots, \beta_n) \in F^n$$

което е решение на всяко от y -ята в системата

a_{ij} - N^o неизвестно
 N^o уравнение

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \dots & a_{kn} \end{pmatrix}_{k \times n} = (a_{ij})_{k \times n}$$

k реда и n колонки

$$\bar{A} = \left(\begin{array}{ccc|c} a_{11} & \dots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{k1} & \dots & a_{kn} & b_k \end{array} \right) - \text{разширена матрица на систем}$$

$$\bar{A} = \left(\begin{array}{ccc|c} 3 & 2 & 1 & 2 \\ 5 & -1 & 2 & 2\alpha \\ 7 & 4 & 0 & 3\alpha \\ 1 & 0 & -1 & 4\alpha \end{array} \right)$$

Етап преобразуване на систем

1. редукция на местата на y -ите

2. униконтиране по скалар $\neq 0$

3. униконтиране и избирателни y -и към други

За матрица $y^{-1} = \text{ref}$

Еквивалентни системи

Две системи са екв когато множествата им от решения са еднакви

несъвместна - няма решение
съвместна - има решение

определена - има единствено решение
неопределена - има повече от 1 решение

*ТВ Елем. преобразование предр една система в еквивалентна на нея

$\bar{A}$ четиря + към 1ва
чети x 2 + към 2ра

$$\bar{A} = \left(\begin{array}{ccc|c} 4 & 2 & 0 & 5\alpha \\ 7 & -1 & 0 & 10\alpha \\ 7 & 4 & 0 & 3\alpha \\ 1 & 0 & -1 & 4\alpha \end{array} \right) \sim \left(\begin{array}{ccc|c} 18 & 0 & 0 & 25\alpha \\ 7 & -1 & 0 & 10\alpha \\ 35 & 0 & 0 & 43\alpha \\ 1 & 0 & -1 & 4\alpha \end{array} \right)$$

$$18\alpha = 25\alpha \quad \alpha = \frac{25}{18}\alpha$$

$$35\alpha = 43\alpha$$

$$\frac{25}{18} 25\alpha = 43\alpha$$

$$\left(43 - \frac{25 \cdot 25}{18} \right) \alpha = 0 \\ = 1\alpha = 0$$

$$\alpha = 0 \Rightarrow \gamma = 0$$

$$\gamma \lambda - \varphi = \gamma 0 \alpha \Rightarrow \gamma \varphi = 0$$

$$\gamma = 0$$

При $\alpha = 0$ вектора ϕ е линейна комбинация.

* линейная оболочка

$$l(a^{(1)}, a^{(k)}) = \{ \lambda_1 a^{(1)} + \dots + \lambda_k a^{(k)} \mid \lambda_1, \dots, \lambda_k \in F \}$$

$$l(a, b, c) = \{ \lambda_1 a + \lambda_2 b + \lambda_3 c \mid \lambda_i \in \mathbb{R} \} = \{ \lambda_1 + 2\lambda_2 + \lambda_3, \lambda_1 - \lambda_2 + 2\lambda_3, 7\lambda_1 + 4\lambda_2 \}$$

$$l(\theta) = \{ \theta \}$$

$$\{ \lambda_1 + 2\lambda_2 + \lambda_3, \lambda_1 - \lambda_2 + 2\lambda_3, 7\lambda_1 + 4\lambda_2 \} \mid \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$$

$$U \subset F^n \quad U \neq \emptyset$$

U е линейно подпространство на F^n когато

$$1 \quad a + b \in U, \quad \forall a, b \in U$$

$$2 \quad \lambda a \in U, \quad \forall a \in U, \quad \forall \lambda \in F$$

$$U \subset F^n$$

$$1 \quad \{0\} \subset F^n$$

$$2 \quad U = \{p, 2p, 3p, 4p, 5p\} \mid p \in \mathbb{Z}\}$$

$$U \subset \mathbb{Z}^5$$

$$* \text{ TB } U \subset F^n$$

$$1 \quad 0 \in U$$

$$a \in U \Rightarrow 0 = 0a \in U$$

$$2 \quad a^{(1)}, a^{(2)} \in U$$

$$\Rightarrow \lambda_1 a^{(1)} + \lambda_2 a^{(2)} + \dots + \lambda_k a^{(k)} \in U$$

$$3 \quad a^{(1)}, a^{(2)} \in U$$

$$(a^{(1)}, a^{(2)}) \subset U$$

$$* \text{ TB } a^{(1)}, a^{(k)} \in F^n$$

$$1 \quad (a^{(1)}, a^{(k)}) \text{ e му погпр-во}$$

$$b = (\lambda_1, a^{(1)}, \dots, \lambda_k, a^{(k)}) \in (a^{(1)}, a^{(k)})$$

$$c = \varphi_1 a^{(1)} + \dots + \varphi_k a^{(k)} \in \text{---}$$

$$b \in c = (\lambda_1 + \varphi_1) a^{(1)} + \dots + (\lambda_k + \varphi_k) a^{(k)} \in \text{---}$$

$$\alpha \in F \Rightarrow \alpha \cdot b \in \text{---}$$

*ТВ Решения на хомогенна система е лн. подпр-во на F^n

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ \vdots \\ a_{k1}x_1 + \dots + a_{kn}x_n = 0 \end{cases} \quad \begin{array}{l} U - \text{лн. во от рещи} \\ U \neq \emptyset \text{ защото } \theta(0, \dots, 0) \text{ е рещи} \\ \theta \in U \end{array}$$

$$\beta = (\beta_1, \dots, \beta_n) \in U$$

$$\gamma = (\gamma_1, \dots, \gamma_n) \in U$$

$$a_{11}\beta_1 + \dots + a_{1n}\beta_n = 0$$

$$a_{11}\gamma_1 + \dots + a_{1n}\gamma_n = 0$$

$$a_{11}(\beta_1 + \gamma_1) + \dots + a_{1n}(\beta_n + \gamma_n) = 0$$

$$\beta + \gamma = (\beta_1 + \gamma_1, \beta_2 + \gamma_2, \dots, \beta_n + \gamma_n) \text{ решение на урав. №:}$$

$$\Rightarrow \beta + \gamma \text{ е рещи на сист.}$$

$$\beta + \gamma \in U$$

$$\lambda \in F$$

$$\lambda\beta \text{ е решение}$$

$$a_{11}(\lambda\beta_1) + \dots + a_{1n}(\lambda\beta_n) = \lambda \beta \in U$$

$$U \leq F^n$$

Когато системата не е хомогенна -

и някъде 1 коеф $\neq 0$ тогава лн. во от рещи не е лн. подпространство

u, W - подпространства на F^n

$$u + W = \{ a + b \mid a \in u, b \in W \}$$

сумата на подпр-ва

*Тв. Ако u, W са подпр-ва

$$\Rightarrow u \cap W \text{ и } u + W$$

са подпр-ва на F^n

$$a, b \in u + W$$

$$a = c^{(1)} + d^{(1)}$$

$$c^{(1)} \in u$$

$$b = c^{(2)} + d^{(2)}$$

$$d^{(1)} \in W$$

$$a + b = (c^{(1)} + d^{(1)}) + (c^{(2)} + d^{(2)}) = \underbrace{(c^{(1)} + c^{(2)})}_{\in u} + \underbrace{(d^{(1)} + d^{(2)})}_{\in W} \in u + W$$

$$\lambda \in F \quad \lambda a = \lambda (c^{(1)} + d^{(1)}) = \underbrace{\lambda c^{(1)}}_{\in u} + \underbrace{\lambda d^{(1)}}_{\in W}$$

$$\lambda a \in u + W$$

линейная зависимость и независимость

$$\{a_1, \dots, a_k\} \subseteq F^n$$

Опр $\{a_1, \dots, a_k\}$ е л. зависима когато $\exists \lambda_1, \dots, \lambda_k \in F$
за които $(\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$

$$\text{такава че } \lambda_1 a_1 + \dots + \lambda_k a_k = 0$$

13

Опр $\{a_1, \dots, a_k\}$ е мин. независ., когато за
 всички $(\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$ $\lambda_i \in F$
 е изпълнено $\lambda_1 a_1 + \dots + \lambda_k a_k \neq 0$ ЛНЗ

$$a = (1, 1, 1) \quad b = (0, 1, 2) \quad c = (30, 25, 20)$$

$$\alpha a + \beta b + \gamma c = 0$$

$$\alpha \neq 0 \quad \beta = 0 \quad \gamma = 0$$

$$\alpha + 0\beta + 30\gamma = 0$$

$$\alpha + \beta + 25\gamma = 0$$

$$\alpha + 2\beta + 20\gamma = 0$$

$$\begin{pmatrix} 1 & 0 & 30 \\ 1 & 1 & 25 \\ 1 & 2 & 20 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 30 \\ 0 & 1 & -5 \\ 0 & 2 & -10 \end{pmatrix} \begin{matrix} (2) \\ \div 2 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 0 & 30 \\ 0 & 1 & -5 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\gamma = p \quad \beta - 5\gamma = 0$$

$$\beta = 5p$$

$$\alpha + 30\gamma = 0 \Rightarrow \alpha = -30p$$

$$p \in \mathbb{R}$$

$$(-30p, 3p, p)$$

$$(-30, 5, 1) \Rightarrow 13$$

$$\{a, b\}$$

$$xa + yb = 0$$

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 2 \end{pmatrix} \quad \begin{matrix} x=0 \\ y=0 \\ 2y=0 \end{matrix} \quad \begin{matrix} a, b \text{ ca} \\ 143 \end{matrix}$$

$$1 \quad \{a\} \subset F^n \quad \{a\} \in 13 \Leftrightarrow a = 0$$

$$a \neq 0 \quad \lambda \neq 0 \quad \lambda a = 0 \text{ caso } a \neq 0$$

$$2 \quad a, b \in F^n \quad a, b \text{ ca } 13 \Leftrightarrow \exists c \text{ pr } \text{proprietat}$$

$$\text{T.e. } a = \alpha b \text{ unde } b = \beta \cdot a$$

$$\text{Uc } a, b \text{ ca } 13 \Rightarrow \exists \lambda_1, \lambda_2 \in F: \lambda_1 + \lambda_2 b = 0 \\ \sim (\lambda_1, \lambda_2) = (0, 0)$$

$$a \neq 0 \quad \lambda_2 \neq 0 \Rightarrow \exists \lambda_2^{-1} \in F$$

$$\lambda_2 b = -\lambda_1 a \quad \lambda_2^{-1} \lambda_2 b = -\lambda_1 \lambda_2^{-1} a \quad b = (-\lambda_1, \lambda_2^{-1}) a$$

Нека a, b са пропорционални

$$a = \alpha b \quad \alpha \in F$$

$$a - \alpha b = 0$$

$$(1, -\alpha) \neq (0, 0)$$

$$\Rightarrow a, b \text{ ЛЗ}$$

* Тв Нека $a_1, \dots, a_k \in F^n \quad k \geq 2$

$\{a_1, \dots, a_k\} \in \text{ЛЗ} \Leftrightarrow$ едн от векторите е
линейна комбинация на останалите

$$\Rightarrow \text{Нека ЛЗ}$$

$$\Rightarrow \exists \lambda_1, \dots, \lambda_k \in F \quad (\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$$

$$\Rightarrow \exists i, \lambda_i \neq 0$$

$$\lambda_1 a_1 + \dots + \lambda_i a_i + \dots + \lambda_k a_k = 0$$

$$\lambda_i a_i = -(\lambda_1 a_1 + \dots + \lambda_k a_k)$$

$$\lambda_i^{-1} \lambda_i a_i = -\lambda_i^{-1} (\lambda_1 a_1 + \dots + \lambda_{i-1} a_{i-1} + \lambda_{i+1} a_{i+1} + \dots + \lambda_k a_k)$$

$$\Leftarrow \text{Нека } a_i \in \langle a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_k \rangle$$

$$\Rightarrow \exists \gamma_j \quad \gamma_j \in (\text{коэф.})$$

$$a_i = \gamma_1 a_1 + \dots + \gamma_{i-1} a_{i-1} + \gamma_{i+1} a_{i+1} + \dots + \gamma_k a_k$$

$$\gamma_1 a_1 + \dots + \gamma_{i-1} a_{i-1} - 1 a_i + \gamma_{i+1} a_{i+1} + \dots + \gamma_k a_k = 0$$

$$(\gamma_1, \dots, \gamma_{i-1}, -1, \gamma_{i+1}, \dots, \gamma_k) \neq (0, \dots, 0) \Rightarrow \text{ЛЗ}$$

* Тв ако $\{a_1, \dots, a_k\} \in F^n$
 $\in L$ тогава $\{a_1, \dots, a_k, a_{k+1}, \dots, a_p\} \in L$
 също $\in L$

Об. 60

$$L \Rightarrow \exists (\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$$

$$\lambda_1 a_1 + \dots + \lambda_k a_k = 0$$

$$\lambda_1 a_1 + \dots + \lambda_k a_k + 0 a_{k+1} + \dots + 0 a_p = 0$$

$$\Rightarrow \{a_1, \dots, a_k, a_{k+1}, \dots, a_p\} \in L$$

Следствие

1 Ако в съвкупност от вектор $тв \in L$
 2 Ако в съвкупност от вектор има 2 еднакви вектора
 3 Ако има 2 пропорционални $v. \Rightarrow \in L$

$$F^n = \{(a_1, a_2, \dots, a_n) \mid a_i \in F\}$$

$$a_1, \dots, a_k \in F^n$$

$a_i = (a_{i1}, \dots, a_{in})$ са $L \Leftrightarrow$ матрица която издържа
 тези вектори

А е матрица на хомогенна система,
 която има ненулево решение

$$\lambda_1 a_1 + \dots + \lambda_k a_k = 0$$

$$\lambda_1 + \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{pmatrix} + \dots + \lambda_k \begin{pmatrix} a_{k1} \\ a_{k2} \\ \vdots \\ a_{kn} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{21}x_2 + \dots + a_{n1}x_n = 0 \\ \vdots \\ a_{1k}x_1 + a_{2k}x_2 + \dots + a_{nk}x_n = 0 \end{cases}$$

има ненулево решение

$$\rightarrow \begin{pmatrix} a_{11}\lambda_1 + \dots + a_{1k}\lambda_k \\ a_{21}\lambda_1 + \dots + a_{2k}\lambda_k \\ \vdots \\ a_{n1}\lambda_1 + \dots + a_{nk}\lambda_k \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Пример $a_1 = (1, 3, 5, 7)$
 $a_2 = (8, 6, 4, 2)$
 $a_3 = (3, 3, 3, 3)$

$$\begin{cases} 1\lambda_1 + 8\lambda_2 + 3\lambda_3 = 0 \\ 3\lambda_1 + 6\lambda_2 + 3\lambda_3 = 0 \\ 5\lambda_1 + 4\lambda_2 + 3\lambda_3 = 0 \\ 7\lambda_1 + 2\lambda_2 + 3\lambda_3 = 0 \end{cases}$$

$$\begin{pmatrix} 1 & 8 & 3 \\ 3 & 6 & 3 \\ 5 & 4 & 3 \\ 7 & 2 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 8 & 3 \\ 2 & 2 & 0 \\ 4 & -4 & 0 \\ 6 & -6 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 8 & 3 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_1 + 8\lambda_2 + 3\lambda_3 = 0$$

$$2\lambda_1 - 2\lambda_2 = 0$$

$$\lambda_2 = p \quad \lambda_1 = p$$

$$p \in (p, p, -3p) \quad \begin{aligned} p + 8p + 3\lambda_3 &= 0 \\ \lambda_3 &= -3p \end{aligned}$$

$$\Rightarrow 13$$

* Тв $a_k \in \mathcal{L}$
 тогава всяко полином-во число $\mathcal{L}$

Лема за $\mathcal{L}$

Има $\{a_1, \dots, a_k\} \subset F^n$

както $\mathcal{L}$ $b \in F^n$

$\{a_1, \dots, a_k, b\} \in \mathcal{L}$

$b \notin \langle a_1, \dots, a_k \rangle$

Д-во $\Rightarrow$

Допускаме $a \in \langle a_1, \dots, a_k \rangle$

$b = \lambda_1 a_1 + \dots + \lambda_k a_k$

$\lambda_1 a_1 + \dots + \lambda_k a_k - b = 0$

$\Rightarrow$ е $\mathcal{L}$ -противоречие

$\Leftarrow b \notin \langle a_1, \dots, a_k \rangle$

Допускаме $\{a_1, \dots, a_k\} \subset \mathcal{L}$

$\exists (\lambda_1, \dots, \lambda_k, \mu) \neq (0, \dots, 0, 0)$

$\lambda_1 a_1 + \dots + \lambda_k a_k + \mu b = 0$

сл 1 $\mu \neq 0 \Rightarrow \exists \mu^{-1} \in F$

$b = -\mu^{-1}(\lambda_1 a_1 + \dots + \lambda_k a_k)$

т.е. $b \in \langle a_1, \dots, a_k \rangle$

$\Rightarrow$ противоречие

сл 2 $\mu = 0$ от $\lambda_1, \dots, \lambda_k, \mu \neq (0, \dots, 0)$

$\Rightarrow (\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$

$\lambda_1 a_1 + \dots + \lambda_k a_k + 0b = 0$

$\lambda_1 a_1 + \dots + \lambda_k a_k = 0$

$\Rightarrow \mathcal{L}$ -противоречие

т.е. $\{a_1, \dots, a_k, b\} \in \mathcal{L}$

* Основная лема из Линейной Алгебры

$$\text{Или } \{a_1, \dots, a_k\} \subset F^n$$

$$\{b_1, \dots, b_s\} \subset F^n$$

$$b_1, \dots, b_s \in \langle a_1, \dots, a_k \rangle$$

$$s \geq k$$

$$\Rightarrow \{b_1, \dots, b_s\} \text{ с. л.з.}$$

Индукция по k

$$\text{при } k=1$$

$$b_1 = \alpha_1 a_1$$

$$b_2 = \alpha_2 a_1$$

$$\text{Или } \alpha_1 = 0 \Rightarrow b_1 = 0 \Rightarrow \{b_1, \dots, b_s\} \text{ с. л.з.}$$

$$\text{Или } \alpha_1 \neq 0 \Rightarrow a_1 = \alpha_1^{-1} b_1 \Rightarrow b_2 = \alpha_2 \alpha_1^{-1} b_1 \Rightarrow \{b_1, \dots, b_s\} \text{ с. л.з.}$$

Допущение: те тв. е верно для $k-1$
и для $\{a_1, \dots, a_k\}, \{b_1, \dots, b_s\}; b_i \in \langle a_1, \dots, a_k \rangle$

$$b_1 = \lambda_{11} a_1 + \dots + \lambda_{1,k-1} a_{k-1} + \lambda_{1k} a_k$$

...

$$b_{s-1} = \lambda_{s-1,1} a_1 + \dots + \lambda_{s-1,k-1} a_{k-1} + \lambda_{s-1,k} a_k$$

$$b_s = \lambda_{s,1} a_1 + \dots + \lambda_{s,k-1} a_{k-1} + \lambda_{sk} a_k$$

$$\text{Или } \lambda_{s,1} = \dots = \lambda_{s,k-1} = \lambda_{sk} = 0 \Rightarrow b_s = 0 \Rightarrow \{b_1, \dots, b_s\} \text{ с. л.з.}$$

2.1 $\exists \lambda_{si} \neq 0$ и т.д. $\lambda_{sk} \neq 0$

$$b_s = \lambda_{s1} a_1 + \dots + \lambda_{s, k-1} a_{k-1} + \lambda_{sk} a_k \quad | \cdot - \frac{\lambda_{1k}}{\lambda_{sk}}$$

$$c_1 = b_1 - \frac{\lambda_{1k}}{\lambda_{sk}} b_s = \lambda'_{11} a_1 + \dots + \lambda'_{1, k-1} a_{k-1} + 0 a_k$$

$$c_{s-1} = b_{s-1} - \frac{\lambda_{s-1, k}}{\lambda_{sk}} b_s = \lambda'_{s-1, 1} a_1 + \dots + \lambda'_{s-1, k-1} a_{k-1} + 0 a_k$$

$$c_1, c_2, \dots, c_{s-1} \in F^n \text{ и } c_i \in \langle a_1, \dots, a_{k-1} \rangle$$

$s-1 \geq k-1$ прилагая индукцию по количеству переменных

$$\Rightarrow c_1, \dots, c_{s-1} \text{ с } \lambda \Rightarrow \exists (\alpha_1, \dots, \alpha_{s-1})$$

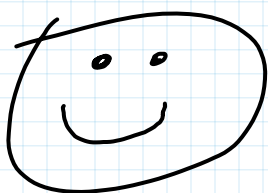
$$\alpha_1 c_1 + \dots + \alpha_{s-1} c_{s-1} = \vec{0} \quad \begin{pmatrix} \neq \\ 0, \dots, 0 \end{pmatrix}$$

$$\alpha_1 \left(b_1 - \frac{\lambda_{1k}}{\lambda_{sk}} b_s \right) + \dots + \alpha_{s-1} \left(b_{s-1} - \frac{\lambda_{s-1, k}}{\lambda_{sk}} b_s \right) = \vec{0}$$

$$\alpha_1 b_1 + \alpha_2 b_2 + \dots + \alpha_{s-1} b_{s-1} - \left(\frac{\alpha_1 \lambda_{1k}}{\lambda_{sk}} + \dots + \frac{\alpha_{s-1} \lambda_{s-1, k}}{\lambda_{sk}} \right) b_s = \vec{0}$$

$$\underbrace{(\alpha_1, \dots, \alpha_{s-1}, \beta)}_{\neq (0, \dots, 0)} \neq (0, \dots, 0)$$

$$\Rightarrow b_1, \dots, b_s \text{ с } \lambda \quad \vec{0} \\ \Rightarrow \tau b \in \text{вектор}$$



$$\{a_1, \dots, a_k\} \subseteq CF^*$$

$$\{a_{i_1}, \dots, a_{i_s}\} \subseteq \{a_1, \dots, a_k\}$$

е максимално много взаимно
поумножимо когато

$$1 \quad \{a_{i_1}, \dots, a_{i_s}\} \in \text{МНЗ}$$

$$2 \quad a_j \in \bigcap_{i=1}^s (a_{i_1}, a_{i_s}) \quad \forall j=1, k$$

$$\{a_{i_1}, \dots, a_{i_s}, a_j\} \in \text{МНЗ}$$

$$* \quad \{a_1, \dots, a_k\} \subseteq CF^*$$

$$\{a_{i_1}, \dots, a_{i_s}\} \in \text{МНЗ}$$

$$\{a_j, a_k\} \in \text{МНЗ}$$

$$\Rightarrow s=k \quad \text{Д.б.} \quad a_{j_1}, a_{j_2} \in (a_{i_1}, a_{i_s})$$

Доказателство че $r \geq s$ от Оси. 1 a_{j_1}, a_{j_2}
Доказателство че $r \leq s$
 $a_{i_1}, a_{i_s} \in (a_{j_1}, a_{j_k})$ Против 13
От Оси. 1 $\Rightarrow a_{i_1}, a_{i_s} \in \text{МНЗ}$ - против

* Опр $\nu(a_1, \dots, a_k) = 1$
 когато ν е бр-я на векторите
 в еф че МЛНЗП на $a_1, \dots, a_k$

Т Нека $\{a_1, \dots, a_k\} \subset F^n$

а) n поне един от вект е $\neq 0$

$\Rightarrow \exists$ МЛНЗП на $\{a_1, \dots, a_k\}$

б) $u = 1(a_1, \dots, a_k)$ (n поне един $\neq 0$)

$\Rightarrow \exists$ МЛНЗП на u
 която е родик
 на $\{a_1, \dots, a_k\}$

в) W е лнк. подпр. в n

В която има поне един ненулев вектор

$\Rightarrow \exists$ МЛНЗП на W

Нека $b_1 \neq 0$ ($\in \{a_1, \dots, a_k\}$ или $b_1 \in W$) $\Rightarrow \{b_1\}$ ЛНЗ

$\{a_1, \dots, a_k\} \subset 1(b_1)$

$\{b_1\}$ МЛНЗП $\exists a_1 \in 1(b_1), b_2 = a_1$ $\{b_1, b_2\}$ ЛНЗ $\exists b_2 \in W, b_2 \neq 1(b_1)$
 $\{a_1, a_2\} \subset 1(b_1, b_2)$ $\{b_1, b_2\}$ МЛНЗП

$\{b_1, b_2\}$ ЛНЗ $\exists b_3 = a_j \notin 1(b_1, b_2) = \{b_1, b_2, b_3\}$

след поемане к стълби $\{b_1, \dots, b_r\}$ МЛНЗП



$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{k1} & \dots & a_{kn} \end{pmatrix} \rightarrow \left. \begin{matrix} a_1(a_{11} \dots a_{1n}) \\ a_2(a_{21} \dots a_{2n}) \\ \vdots \\ a_k(a_{k1} \dots a_{kn}) \end{matrix} \right\} \in F^n$$

$$C_1 = \begin{pmatrix} a_{11} \\ \vdots \\ a_{k1} \end{pmatrix}, C_2 = \begin{pmatrix} a_{12} \\ \vdots \\ a_{k2} \end{pmatrix}, \dots, C_n = \begin{pmatrix} a_{1n} \\ \vdots \\ a_{kn} \end{pmatrix} \in F^k$$

Елем. преобр. на ред

- 1 умножаване по скалар $\neq 0$
- 2 размяна на места на два реда
- 3 прибавяне на един ред към друг ред умножен по скалар

$$A \sim A_1 \sim A_2 \dots \sim R$$

* Теорема Елементарните преобразовани по редове на една матрица не променят ранга на съкупността от вектори и минималната обвивка на редовете на матрицата.
 $A \sim A'$ редове на A са $a_1, a_2, \dots, a_k$ с мин. обв. и образув. матрици
 — „ — A' са $a'_1, a'_2, \dots, a'_k$

$$r(a_1 \dots a_k) = r(a'_1 \dots a'_k)$$

$$\exists a_i = \lambda a_i, (\lambda \in F; \lambda \neq 0)$$

$$a_j = a_j \text{ за } j \neq i$$

$$a_i \in l(a_1 \dots a_k)$$

$$l(a_1 \dots a'_k) \subset l(a_1 \dots a_k)$$

$$a_i = \lambda a_i \xrightarrow{\lambda \neq 0} \lambda \neq 0 \Rightarrow \lambda \in F$$

$$a_i = \lambda^{-1} a'_i$$

$$\{a_1 \dots a_k\} \subset l(a'_1 \dots a'_k) \Rightarrow l(a_1 \dots a_k) = l(a'_1 \dots a'_k)$$

$$2 \quad a'_i = a_j; \quad a'_j = a_i, \quad (a'_s = a_s \text{ когато } s \neq \{i, j\})$$

$$\{a_1, \dots, a_k\} = \{a'_1, \dots, a'_k\} \Rightarrow r(a_1, \dots, a_k) = r(a'_1, \dots, a'_k)$$

$$3 \quad a'_i = a_i + \lambda a_j, \quad (i \neq j) \quad a'_s = a_s, \quad \text{за } s \neq i$$

$$a_i \in (a_1, \dots, a_k) \rightarrow l(a'_1, \dots, a'_k) \subset l(a_1, \dots, a_k)$$

$$a'_i = a_i + \lambda a_j \Rightarrow a_i = a'_i - \lambda a_j = a'_i - \lambda a'_j$$

$$\Rightarrow l(a_1, \dots, a_k) = l(a'_1, \dots, a'_k)$$

$$\Rightarrow r(a_1, \dots, a_k) = r(a'_1, \dots, a'_k)$$

* Теорема Елементарните преобразования по редове на една матрица не променят ранга на съвкупността от вектор-столбове

* Лема Нека $A \in M_{k \times n}(F)$ има столбове $c_1, \dots, c_n$ тогава $c_1, \dots, c_k$ са л.з. $\Leftrightarrow$ хомогенната система с матрица A има ненулево решение

$$c_1, \dots, c_n \text{ са л.з. когато } \exists \lambda_1, \dots, \lambda_n \in F \quad (\lambda_1, \dots, \lambda_n) \neq (0, \dots, 0) \quad \lambda_1 c_1 + \dots + \lambda_n c_n = 0$$

$$\lambda \begin{pmatrix} a_{11} \\ \vdots \\ a_{k1} \end{pmatrix} + \dots + \lambda_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{kn} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11}\lambda_1 + \dots + a_{1n}\lambda_n \\ \vdots \\ a_{k1}\lambda_1 + \dots + a_{kn}\lambda_n \end{pmatrix}$$

$$\begin{cases} a_{11}\lambda_1 + \dots + a_{1n}\lambda_n = 0 \\ \vdots \\ a_{k1}\lambda_1 + \dots + a_{kn}\lambda_n = 0 \end{cases} \Leftrightarrow (\lambda_1, \dots, \lambda_n) \in \text{ядро на}$$

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ \vdots \\ a_{k1}x_1 + \dots + a_{kn}x_n = 0 \end{cases}$$

* Лема Елем. преобразования привенят системата във еквивалентна на нея

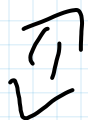
Нека A' е получена чрез ел. преобр. на редовете

$$1 \begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ \vdots \\ a_{k1}x_1 + \dots + a_{kn}x_n = 0 \end{cases} \quad \text{и} \quad 2 \begin{cases} a'_{11}x_1 + \dots + a'_{1n}x_n = 0 \\ \vdots \\ a'_{k1}x_1 + \dots + a'_{kn}x_n = 0 \end{cases}$$

двете системи са еквивалентни

$c_1, \dots, c_r$ са МЛНЗ на $\{c_1, \dots, c_n\}$

$\Rightarrow c_1, \dots, c_r$ ЛНЗ



$$\begin{cases} a_{11}x_1 + \dots + a_{1r}x_r = 0 \\ \vdots \\ a_{k1}x_1 + \dots + a_{kr}x_r = 0 \end{cases}$$

и са само нулево р-и.

$$\Leftrightarrow \begin{cases} a'_{11}x_1 + \dots + a'_{1r}x_r = 0 \\ \vdots \\ a'_{k1}x_1 + \dots + a'_{kr}x_r = 0 \end{cases}$$

и $\{c_1, \dots, c_r, c_i\} \in \text{ЛНЗ} \Leftrightarrow \{c'_1, \dots, c'_r, c'_i\} \in \text{ЛНЗ}$
 $\forall i \in \{1, \dots, k\}$

$\forall j \in \{1, \dots, k\}$

$c'_1, \dots, c'_r$

са МЛНЗ

и са стълбовете

на A'



$$r(c'_1, \dots, c'_n) = r$$

$$A \in M_{n \times n}(F)$$

$$A^T = A^t = B \in M_{n \times n}(F) \quad B = (b_{ij})$$

$$b_{ij} = a_{ji}$$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} \rightarrow A^t = \begin{pmatrix} 1 & 4 & 7 & 10 \\ 2 & 5 & 8 & 11 \\ 3 & 6 & 9 & 12 \end{pmatrix}$$

* С1 Еlemen. преобр. по стълбове не променят реда на съвкупност от вектор стълбове

С12 Еlemen. преобр. на стълбове не променят реда на съвкупност от вектор реда

* Тeорема Нека $A \in M_{n \times n}(F)$ е ненулева матрица
извежда по от еlemen. преобр. по реда и по стълбове
такава че A да се приведе до скаляр вид

$$R = \begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & & & & \\ 0 & & \ddots & & & \\ 0 & & & 1 & & \\ \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix} \text{ Ако } b \text{ в } R \text{ има } 1 \text{ съответно } \Rightarrow \text{ (редове на } A) = \text{ (стълбове на } A)$$

$$\begin{pmatrix} x \end{pmatrix} \sim \begin{pmatrix} x \end{pmatrix} \sim \begin{pmatrix} x \end{pmatrix} \sim \begin{pmatrix} 1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & & & & \\ & A_1 & & & \\ 0 & & & & \end{pmatrix} \quad \begin{aligned} & \text{ (редове на } A) = \text{ (редове на } A') \\ & \text{ (стълбове на } A) = \text{ (стълбове на } A') \end{aligned}$$

$$\text{Ако } k=1 \text{ получим сме } (1 \ 0 \dots 0) A_1, \quad A_1 = 0$$

$$n=1 \quad \text{---} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & & \\ \vdots & & \end{pmatrix}$$



Индукция по матрице?
?

* Т. Руне

$$\text{система } 1 \Leftrightarrow r(A) = r(\bar{A})$$

Т нека H е подр-во от R -я на хомогенната (1_a)

$$\dim H = n - r(A) = m$$

Нека $y_1, \dots, y_m$ е ФЛР на (1_b)

$$H = \{ (y_1, \dots, y_m) = \{ p_1 y_1 + \dots + p_m y_m \mid p_i \in E \}$$

параметри

СЛ (1) е ли система n -броя на неизвестните

1 $r(A) \neq r(\bar{A}) \Leftrightarrow$ неизвестна система

2 $r(A) = r(\bar{A}) = n \Leftrightarrow (1)$ има единствено решение т.е. системата е определена

3 $r(A) = r(\bar{A}) < n \Leftrightarrow (1)$ е неопределена и $m = n - r(A)$

$$\{ x + p_1 y_1 + \dots + p_m y_m \mid p_i$$

$$y_1, \dots, y_m \in \text{ФЛР}$$

и E параметри
или n -ответната хомогенна система

* Т нека U ли подр-во на F^n съществувва хомогенна система с n неизвестни чито решения с U

$$(c_1, \dots, c_n) \in F^n \text{ е решение на } a_1 x_1 + \dots + a_n x_n = 0$$



$$a_1 c_1 + \dots + a_n c_n = 0$$

$$(a_1, \dots, a_n) \in F^n$$

е решение на

$$c_1 x_1 + \dots + c_n x_n = 0$$

D -то U -мн подпросто на F^n

$$I_U U = \{0\}$$

За $b_1, \dots, b_k \in F^n$ са базис на U

$$b_i = (b_{i1}, b_{i2}, \dots, b_{in})$$

$$a_1 x_1 + \dots + a_n x_n = 0$$

$$\begin{array}{l} b_1 - \text{row.} \\ \vdots \\ b_k - \text{row.} \end{array} \left\{ \begin{array}{l} a_1 b_{11} + \dots + a_n b_{1n} = 0 \\ \vdots \\ a_1 b_{k1} + \dots + a_n b_{kn} = 0 \end{array} \right.$$

$$(A, I) \left\{ \begin{array}{l} b_{11} x_1 + \dots + b_{1n} x_n = 0 \\ \vdots \\ b_{k1} x_1 + \dots + b_{kn} x_n = 0 \end{array} \right.$$

$$B \left(\begin{array}{ccc} b_{11} & \dots & b_{1n} \\ \vdots & & \vdots \\ b_{k1} & \dots & b_{kn} \end{array} \right)$$

$$r(B) = k$$

k подпр. во е решени на (A, I)

$$\dim U = n - r(B) = n - k = s$$

$$\alpha_1, \dots, \alpha_s \in \mathbb{C}^n$$

базис на U

$$\alpha_j = (\alpha_{j1}, \dots, \alpha_{jn})$$

$$(A_2) \begin{cases} \alpha_{11}x_1 + \dots + \alpha_{1n}x_n = 0 \\ \vdots \\ \alpha_{s1}x_1 + \dots + \alpha_{sn}x_n = 0 \end{cases}$$

$$i = 1, \dots, k \quad j = 1, \dots, s$$

$$\alpha_{j1}b_{i1} + \dots + \alpha_{jn}b_{in} = 0$$

$$\Rightarrow \alpha_i \text{ e pecm na } (A_1)$$

$$\Rightarrow b_i \text{ e pecm. na } (A_2) \Rightarrow b_i \in T$$

$$b_1, \dots, b_k \in T$$

$$U = \{(b_1, \dots, b_k) \in T$$

$$\dim U = k \quad \dim T = k$$

$$\Rightarrow U = T \Rightarrow U \text{ e pecmena na cctepaTa } (A_2)$$

$$b_1 = (1, 2, 3, 4, 5)$$

$$b_2 = (6, 7, 8, 9, 10)$$

$$U = \langle b_1, b_2 \rangle$$

$$\alpha_1 x_1 + \dots + \alpha_5 x_5 = 0$$

$$A_1 \begin{cases} x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 0 \\ 6x_1 + 7x_2 + 8x_3 + 9x_4 + 10x_5 = 0 \end{cases}$$

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 5 & 5 & 5 & 5 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned} x_2 &= 2x_3 - 3x_4 - 4x_5 \sim \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 0 & -1 & -1 & -3 \end{pmatrix} \\ x_1 &= x_3 + 2x_4 + 3x_5 \end{aligned}$$

и на мн-во от pecm T

$$A = \begin{pmatrix} \alpha_{11} & \dots & \alpha_{1n} \\ \vdots & & \vdots \\ \alpha_{s1} & \dots & \alpha_{sn} \end{pmatrix}$$

$$r(A) = s$$

$$\dim T = n - s = 4 - (4 - k) = k$$

$$r(B) = 2$$

и нагласно P -е на $(1,)$

$$\dim H = 5 - 2 = 3$$

$\alpha_1, \alpha_2, \alpha_3 \in \mathbb{C}P$

$$\begin{array}{c|ccc} x_1, x_2 & x_3 & x_4 & x_5 \\ \hline (1, -2,) & (1, 0, 0) = \alpha_1 \\ (2, -3,) & (0, 1, 0) = \alpha_2 \\ (2, -4,) & (0, 0, 1) = \alpha_3 \end{array}$$

$\mathbb{C}P$ на $(1,)$

$$\begin{array}{l|l} \Lambda_2 & x_1 - 2x_2 + x_3 = 0 \\ & 2x_1 - 3x_2 + x_4 = 0 \\ & 3x_1 - 4x_2 + x_5 = 0 \end{array}$$

U е решението на Λ_2

~~X~~ Среда в 17.00
Квартал 20.00

$M_{K \times n}(F)$

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & \ddots & \vdots \\ a_{k1} & \dots & a_{kn} \end{pmatrix} = (a_{ij})_{K \times n}$$

$$B = (b_{ij})_{k \times n} \in M_{k \times n}(\bar{F})$$

$$A+B = \begin{pmatrix} a_{11}+b_{11} & \dots & a_{1n}+b_{1n} \\ \vdots & \ddots & \vdots \\ a_{k1}+b_{k1} & \dots & a_{kn}+b_{kn} \end{pmatrix} \in M_{k \times n}(F)$$

$$\begin{pmatrix} 2 & 3 & 7 \\ 0 & 7 & 4 \end{pmatrix} + \begin{pmatrix} 3 & 5 & 1 \\ 2 & 1 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 8 & 2 \\ 2 & 0 & 8 \end{pmatrix}$$

6-6a $A, B, C \in M_{k \times n}(F)$

1 $A+B = B+A$

2 $(A+B)+C = A+(B+C)$

3 $O = \begin{pmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{pmatrix} \Rightarrow A+O = A \quad \forall A \in M_{k \times n}(F)$

4 $-A = \begin{pmatrix} -a_{11} & -a_{12} & \dots & -a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{k1} & -a_{k2} & \dots & -a_{kn} \end{pmatrix}$

5 $A+(-A) = O$
 $A+X = O$

$X = O+(-A) = -A$

$\lambda \in F \quad A \in M_{k \times n}(F)$

$$\lambda A = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \dots & \lambda a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{k1} & \lambda a_{k2} & \dots & \lambda a_{kn} \end{pmatrix} \in M_{k \times n}(F)$$

6 $O \cdot A = O, \quad 1 \cdot A = A, \quad -1 \cdot A = -A$

7 $\lambda(A+B) = \lambda A + \lambda B \quad \lambda(\mu A) = (\lambda\mu)A = \mu(\lambda A)$

8 $(\lambda\mu)A = \lambda(\mu A) = \mu(\lambda A)$

$$A \in M_{k \times n}(F) \Rightarrow A^t = M_{n \times k}(F)$$

$$a_{ij} \rightarrow j i \in A^t$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^t = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$$10) (A^t)^t = A$$

$$11) (A+B)^t = A^t + B^t$$

$$12) (\lambda A)^t = \lambda A^t$$

$$A_{k \times n} \cdot B_{n \times s} = C_{k \times s}$$

$$a_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 4 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 & 1 \cdot (-1) + 2 \cdot 4 + 3 \cdot 5 \\ -1 \cdot 1 + 4 \cdot 2 + 0 \cdot 3 & -1 \cdot (-1) + 4 \cdot 4 + 0 \cdot 5 \end{pmatrix}$$

$$= \begin{pmatrix} 14 & 22 \\ 7 & 17 \end{pmatrix}$$

Об. 7 A, B матрицы и элемент $\lambda \in F$

$$\lambda(AB) = (\lambda A)B = A(\lambda B)$$

$$1) \lambda(a_{i1}b_{1j} + \dots + a_{in}b_{nj})$$

$$(\lambda a_{i1})b_{1j} + \dots + \lambda a_{in}b_{nj}$$

$$a_{i1}(\lambda b_{1j}) + \dots + a_{in}(\lambda b_{nj})$$

$$2 \quad B', B'' \in M_{n \times s}(F), \quad A \in M_{k \times n}(F)$$

$$A \cdot (B' + B'') = A B' + A B''$$

" "
C

$$C_{ij} = a_{i1}(b'_{1j} + b''_{1j}) + \dots + a_{in}(b'_{nj} + b''_{nj})$$

$$= a_{i1}b'_{1j} + \dots + a_{in}b'_{nj} + a_{i1}b''_{1j} + \dots + a_{in}b''_{nj}$$

на каждо i, j
 $B' + B''$

$$(A' + A'')B = A'B + A''B$$

$$3 \quad A_{k \times n} \Rightarrow A^t \in M_{n \times k}(F)$$

$$B_{n \times s} \Rightarrow B^t \in M_{s \times n}(F)$$

$$C_{ij} = a_{i1}b_{1j} + \dots + a_{in}b_{nj} = (a_{i1} \ a_{i2} \ \dots \ a_{in}) \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix}$$

$$= (b_{1j} \ \dots \ b_{nj}) \begin{pmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{pmatrix} = d_{ji}$$

по j на B^t — строка, на A^t

$$B^t A^t = D \in M_{s \times k}(F)$$

$$D^t = AB \quad (B^t A^t)^t = AB \quad B^t A^t = (AB)^t$$

$$* T \quad A_{k \times n}; B_{n \times s}; C_{s \times t}$$

$$\Rightarrow (AB)C = A(BC)$$

$$AB = D_{k \times s} \quad (AB)C = G_{k \times t}$$

$$BC = L_{n \times t} \quad A(BC) = H_{k \times t}$$

$$g_{ij} = \sum_{p=1}^s c_{ip} c_{pj} = \sum_{p=1}^s \left(\sum_{q=1}^n a_{iq} b_{qp} \right) c_{pj} = \sum_{p=1}^s \sum_{q=1}^n a_{iq} b_{qp} c_{pj}$$

$$h_{ij} = \sum_{q=1}^n a_{iq} l_{qj} = \sum_{q=1}^n a_{iq} \left(\sum_{p=1}^s b_{qp} c_{pj} \right) = \sum_{q=1}^n \sum_{p=1}^s a_{iq} b_{qp} c_{pj} =$$

$$\Rightarrow g_{ij} = h_{ij}$$

$$\Rightarrow G = H$$

$$A \in M_{k \times n}$$

A, A само при $k=n$ होते

Ch. $A \in M_{n \times n} (F)$ $m \in \mathbb{N}$

$$A^m = \underbrace{A \cdot A \cdot A \cdots A}_m$$

$$A^{m+t} = A^m \cdot A^t$$

$$(A^m)^t = A^{m \cdot t} = \underbrace{A \cdot A \cdots A}_t = A^{m \cdot t}$$

~~! $(A^m)^n = A^{m \cdot n}$~~

Ex. $E_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & \cdots & 1 \end{pmatrix} \quad c_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

$$A \in M_{k \times n}$$

$$E_k \cdot A = A$$

$$c_{ij} = e_{i1} a_{1j} + \dots + e_{ii} a_{ij} + \dots + e_{ik} a_{kj}$$

$$c_{ij} = a_{ij} \quad \text{अतः} \quad AE_n = A$$

$$3a \quad A \in M_{n \times n}(F)$$

$$f \in F[x]$$

$$f(x) = b_k x^k + b_{k-1} x^{k-1} + \dots + b_1 x + b_0$$

$$f(A) = b_k A^k + b_{k-1} A^{k-1} + \dots + b_1 A + \underline{b_0 E}$$

$$A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \quad A^2 = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} 9+5 & 3+2 \\ 15+10 & 5+4 \end{pmatrix} = \begin{pmatrix} 14 & 5 \\ 25 & 9 \end{pmatrix}$$

$$f = x^2 - 5x + 1$$

$$\begin{aligned} f(A) &= A^2 - 5A + E = \begin{pmatrix} 14 & 5 \\ 25 & 9 \end{pmatrix} - 5 \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \\ &= \begin{pmatrix} 14 & 5 \\ 25 & 9 \end{pmatrix} - \begin{pmatrix} 15 & 5 \\ 25 & 10 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$